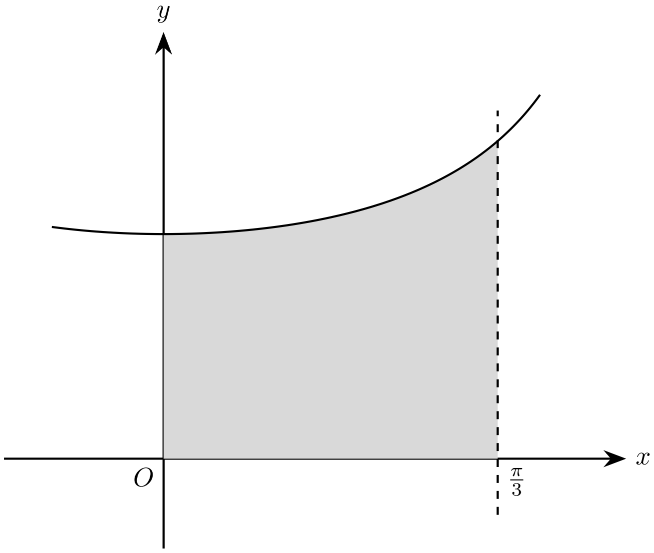
# Questions

**1.** **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(\sec x + \tan x) = \sec x$$ **[2]**




The shaded region is enclosed by the curve $y = \sqrt{\sec x}$, the coordinate axes and the line $x = \frac{\pi}{3}$.

The region is rotated completely about the $x$-axis.

**(b)** Find the exact volume of the solid generated. **[3]**

**Solution**

**(a)** Let $u = \sec x + \tan x$. Differentiating,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sec x \tan x + \sec^2 x = \sec x(\tan x + \sec x) = u \sec x$$

Hence, by the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(\sec x + \tan x) = \frac{1}{u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u}\cdot u\sec x = \sec x$$

as required.

**(b)** Since the region is rotated about the $x$-axis, the volume is

$$V = \pi\int_a^b y^2\,\mathrm{d}x$$

Here the limits are $x = 0$ to $x = \frac{\pi}{3}$, and

$$y = \sqrt{\sec x} \implies y^2 = \sec x$$

So

$$V = \pi\int_0^{\pi/3} \sec x\,\mathrm{d}x$$

From part **(a)**,

$$\int \sec x\,\mathrm{d}x = \ln|\sec x + \tan x| + C$$

so

$$V = \pi \left[\ln(\sec x + \tan x)\right]_0^{\pi/3}$$

Now substitute the limits:

$$V = \pi \left(\ln\left(\sec\frac{\pi}{3} + \tan\frac{\pi}{3}\right) - \ln(\sec 0 + \tan 0)\right)$$

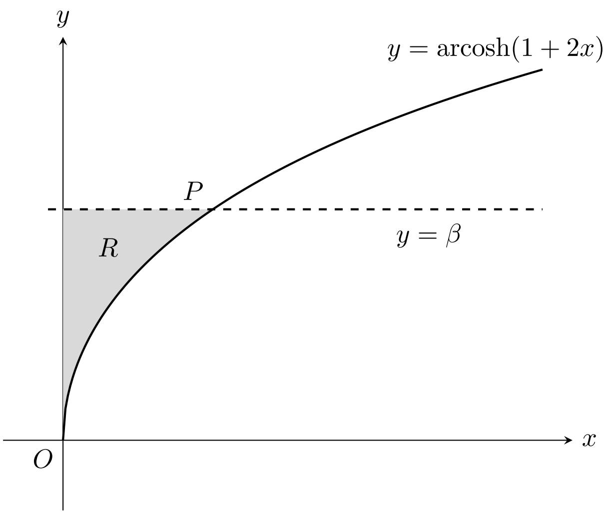
$$\sec\frac{\pi}{3} = 2, \qquad \tan\frac{\pi}{3} = \sqrt{3}, \qquad \sec 0 = 1, \qquad \tan 0 = 0$$

Hence

$$V = \pi \left(\ln(2+\sqrt{3}) - \ln 1\right)$$

$$V = \pi \ln(2+\sqrt{3})$$

So the exact volume is $\pi \ln(2+\sqrt{3})$.




**2.** The diagram above shows a sketch of part of the curve with equation

$$y = \operatorname{arcosh}(1+2x) \qquad x \geq 0$$

and the straight line with equation $y = \beta$.

The line and the curve intersect at the point with coordinates $(\alpha, \beta)$.

Given that $\beta = \ln(2+\sqrt{3})$,

**(a)** show that $\alpha = \frac{1}{2}$. **[3]**

The finite region $R$, shown shaded in the diagram above, is bounded by the curve with equation $y = \operatorname{arcosh}(1+2x)$, the $y$-axis and the line with equation $y = \beta$.

The region $R$ is rotated through $2\pi$ radians about the $y$-axis.

**(b)** Use calculus to find the exact value of the volume of the solid generated. **[6]**

**Solution**

**(a)** At the point of intersection, $(\alpha, \beta)$ lies on the curve

$$y = \operatorname{arcosh}(1+2x)$$

so

$$\beta = \operatorname{arcosh}(1+2\alpha)$$

Applying cosh to both sides gives

$$\cosh\beta = 1+2\alpha$$

Given

$$\beta = \ln(2+\sqrt{3})$$

and using

$$\operatorname{arcosh} 2 = \ln(2+\sqrt{2^2-1}) = \ln(2+\sqrt{3})$$

we have

$$\beta = \operatorname{arcosh} 2$$

so

$$\cosh\beta = 2$$

Therefore
$$1 + 2\alpha = 2$$
and hence
$$2\alpha = 1$$
so
$$\alpha = \frac{1}{2}$$
So the required value is $\alpha = \frac{1}{2}$.

**(b)** First write $x$ in terms of $y$.

From
$$y = \operatorname{arcosh}(1 + 2x)$$
we get
$$\cosh y = 1 + 2x$$
so
$$x = \frac{\cosh y - 1}{2}$$
When $x = 0$,
$$y = \operatorname{arcosh}(1) = 0$$
and the top of the region is $y = \beta$. Hence $0 \le y \le \beta$.

When the region is rotated about the $y$-axis, the volume is
$$V = \pi \int_0^\beta x^2 \,\mathrm{d}y$$
So
$$V = \pi \int_0^\beta \left(\frac{\cosh y - 1}{2}\right)^2 \mathrm{d}y = \frac{\pi}{4} \int_0^\beta (\cosh y - 1)^2 \,\mathrm{d}y$$
Now expand:
$$(\cosh y - 1)^2 = \cosh^2 y - 2\cosh y + 1$$
and use
$$\cosh^2 y = \frac{\cosh 2y + 1}{2}$$
to get
$$\begin{aligned}(\cosh y - 1)^2 &= \frac{\cosh 2y + 1}{2} - 2\cosh y + 1 \\ &= \frac{1}{2}\cosh 2y - 2\cosh y + \frac{3}{2}\end{aligned}$$
Therefore
$$\begin{aligned}V &= \frac{\pi}{4} \int_0^\beta \left(\frac{1}{2}\cosh 2y - 2\cosh y + \frac{3}{2}\right) \mathrm{d}y \\ &= \frac{\pi}{4} \left[\frac{1}{4}\sinh 2y - 2\sinh y + \frac{3}{2}y\right]_0^\beta\end{aligned}$$
Now use the result from part (a), or equivalently $\beta = \operatorname{arcosh} 2$, so
$$\cosh \beta = 2$$
Hence
$$\sinh \beta = \sqrt{\cosh^2 \beta - 1} = \sqrt{4 - 1} = \sqrt{3}$$
since $\beta > 0$.

Also,

$$\sinh 2\beta = 2\sinh\beta\cosh\beta = 2(\sqrt{3})(2) = 4\sqrt{3}$$

Substituting:

$$\begin{aligned} V &= \frac{\pi}{4}\left[\frac{1}{4}(4\sqrt{3}) - 2(\sqrt{3}) + \frac{3}{2}\beta\right] \\ &= \frac{\pi}{4}\left(\sqrt{3} - 2\sqrt{3} + \frac{3}{2}\beta\right) \\ &= \frac{\pi}{4}\left(-\sqrt{3} + \frac{3}{2}\beta\right) \\ &= \frac{\pi}{8}(3\beta - 2\sqrt{3}) \end{aligned}$$

Finally, since $\beta = \ln(2+\sqrt{3})$,

$$V = \frac{\pi}{8}\left(3\ln(2+\sqrt{3}) - 2\sqrt{3}\right)$$

So the exact volume is

$$\frac{\pi}{8}\left(3\ln(2+\sqrt{3}) - 2\sqrt{3}\right)$$

**3.** The equation of a curve is

$$y = \sqrt{\frac{k-x}{k+x}}$$

where $k$ is a positive constant. The region enclosed by the curve, the $x$-axis, the $y$-axis and the line $x = k$ is rotated through $2\pi$ radians about the $x$-axis.

Given that the volume of the solid of revolution formed is 1 unit$^3$, find the exact value of $k$. **[4]**

**Solution**

When the region is rotated about the $x$-axis, the volume is given by

$$V = \pi \int_0^k y^2 \, \mathrm{d}x$$

Here

$$y = \sqrt{\frac{k-x}{k+x}}$$

so

$$y^2 = \frac{k-x}{k+x}$$

Given that the volume is 1 unit$^3$,

$$1 = \pi \int_0^k \frac{k-x}{k+x} \, \mathrm{d}x$$

First simplify the integrand:

$$\frac{k-x}{k+x} = \frac{-(k+x)+2k}{k+x} = -1 + \frac{2k}{k+x}$$

So

$$\begin{aligned} 1 &= \pi \int_0^k \left(-1 + \frac{2k}{k+x}\right) \mathrm{d}x \\ &= \pi \left[-x + 2k\ln(k+x)\right]_0^k \\ &= \pi \left([-k + 2k\ln(2k)] - [0 + 2k\ln k]\right) \\ &= \pi \left(-k + 2k\big(\ln(2k) - \ln k\big)\right) \\ &= \pi \left(-k + 2k\ln 2\right) \\ &= \pi k(2\ln 2 - 1) \end{aligned}$$

Hence

$$k = \frac{1}{\pi(2\ln 2 - 1)}$$

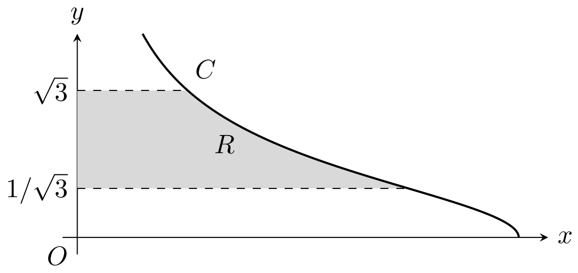
Therefore, the exact value of $k$ is

$$\frac{1}{\pi(2\ln 2 - 1)}$$

**4.** **(a)** Find

$$\int \sin^2\theta \, \mathrm{d}\theta$$

**[2]**




The diagram shows part of the curve $C$ with parametric equations $x = 6\sin^2\theta$, $y = \cot\theta$, $0 < \theta \leq \frac{\pi}{2}$.

The finite region $R$ shown in the diagram is bounded by $C$, the lines $y = \dfrac{1}{\sqrt{3}}$, $y = \sqrt{3}$ and the $y$-axis. Region $R$ is rotated through $2\pi$ radians about the $y$-axis to form a solid of revolution.

**(b)** Show that the volume of the solid of revolution formed is given by the integral

$$k\int_a^b \sin^2\theta \, \mathrm{d}\theta$$

where $a$, $b$ and $k$ are constants to be found. **[5]**

**(c)** Hence find the exact value of this volume, giving your answer in the form $p\pi^2$, where $p$ is a constant to be found. **[3]**

**Solution**

**(a)** Use the identity

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

Then

$$\begin{aligned}\int \sin^2\theta \, \mathrm{d}\theta &= \int \frac{1-\cos 2\theta}{2} \, \mathrm{d}\theta \\ &= \frac{1}{2}\int 1 \, \mathrm{d}\theta - \frac{1}{2}\int \cos 2\theta \, \mathrm{d}\theta \\ &= \frac{\theta}{2} - \frac{1}{2}\cdot\frac{\sin 2\theta}{2} + C \\ &= \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C\end{aligned}$$

Hence

$$\int \sin^2\theta \, \mathrm{d}\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C$$

**(b)** When the region is rotated about the $y$-axis, the volume is

$$V = \pi\int x^2 \, \mathrm{d}y$$

Here

$$x = 6\sin^2\theta, \qquad y = \cot\theta$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = -\operatorname{cosec}^2\theta$$

We now find the $\theta$-limits.

When $y = \sqrt{3}$,

$$\cot\theta = \sqrt{3} \implies \tan\theta = \frac{1}{\sqrt{3}}$$

so $\theta = \dfrac{\pi}{6}$.

When $y = \dfrac{1}{\sqrt{3}}$,

$$\cot\theta = \frac{1}{\sqrt{3}} \implies \tan\theta = \sqrt{3}$$

so $\theta = \dfrac{\pi}{3}$.

Therefore

$$\begin{aligned} V &= \pi \int_{1/\sqrt{3}}^{\sqrt{3}} x^2 \,\mathrm{d}y \\ &= \pi \int_{\pi/3}^{\pi/6} (6\sin^2\theta)^2 \frac{\mathrm{d}y}{\mathrm{d}\theta} \,\mathrm{d}\theta \\ &= \pi \int_{\pi/3}^{\pi/6} (6\sin^2\theta)^2 (-\operatorname{cosec}^2\theta) \,\mathrm{d}\theta \\ &= 36\pi \int_{\pi/3}^{\pi/6} -\sin^2\theta \,\mathrm{d}\theta \\ &= 36\pi \int_{\pi/6}^{\pi/3} \sin^2\theta \,\mathrm{d}\theta \end{aligned}$$

So the required form is

$$k\int_a^b \sin^2\theta \,\mathrm{d}\theta$$

with

$$k = 36\pi, \qquad a = \frac{\pi}{6}, \qquad b = \frac{\pi}{3}$$

**(c)** Using parts (a) and (b),

$$\begin{aligned} V &= 36\pi \int_{\pi/6}^{\pi/3} \sin^2\theta \,\mathrm{d}\theta \\ &= 36\pi \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_{\pi/6}^{\pi/3} \\ &= 36\pi \left(\frac{\pi}{6} - \frac{\sin(2\pi/3)}{4} - \frac{\pi}{12} + \frac{\sin(\pi/3)}{4}\right) \end{aligned}$$

Since

$$\sin\frac{2\pi}{3} = \sin\frac{\pi}{3}$$

the sine terms cancel, giving

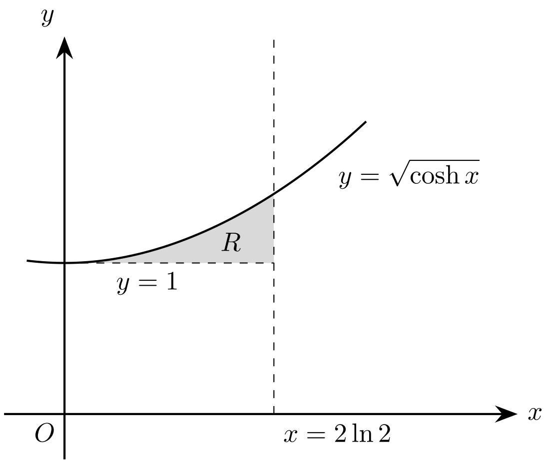
$$\begin{aligned} V &= 36\pi\left(\frac{\pi}{6} - \frac{\pi}{12}\right) \\ &= 36\pi \cdot \frac{\pi}{12} \\ &= 3\pi^2 \end{aligned}$$

Hence the exact volume is

$$3\pi^2$$

so $p = 3$.

**5.** **(a)** Show that $\sinh(2\ln 2) = \dfrac{15}{8}$ **[2]**




The region $R$ is bounded by the curve with equation $y = \sqrt{\cosh x}$, the line $y = 1$ and the line $x = 2\ln 2$, as shown in the diagram. The units of the axes are centimetres.

A designer models a hollow glass ornament as the solid formed by rotating $R$ completely about the $x$-axis.

**(b)** Determine, according to the model, the exact volume of the ornament. **[4]**

**Solution**

**(a)** Using

$$\sinh u = \frac{e^u - e^{-u}}{2}$$

with $u = 2\ln 2$,

$$\begin{aligned}\sinh(2\ln 2) &= \frac{e^{2\ln 2} - e^{-2\ln 2}}{2} \\ &= \frac{e^{\ln 4} - e^{-\ln 4}}{2} \\ &= \frac{4 - \frac{1}{4}}{2} \\ &= \frac{\frac{16}{4} - \frac{1}{4}}{2} = \frac{15/4}{2} = \frac{15}{8}\end{aligned}$$

Hence,

$$\sinh(2\ln 2) = \frac{15}{8}$$

**(b)** First find the left-hand limit of the region by solving where the curve meets $y = 1$:

$$\begin{aligned}\sqrt{\cosh x} &= 1 \\ \cosh x &= 1\end{aligned}$$

and this gives $x = 0$.
So the region extends from $x = 0$ to $x = 2\ln 2$.
When the region is rotated about the $x$-axis, the volume is

$$V = \pi \int_a^b \left((\text{upper curve})^2 - (\text{lower curve})^2\right)\,\mathrm{d}x$$

Here, the upper curve is $y = \sqrt{\cosh x}$ and the lower curve is $y = 1$, so

$$\begin{aligned} V &= \pi \int_0^{2\ln 2} \left( \left( \sqrt{\cosh x} \right)^2 - 1^2 \right) \, \mathrm{d}x \\ &= \pi \int_0^{2\ln 2} (\cosh x - 1) \, \mathrm{d}x \end{aligned}$$

Integrating,

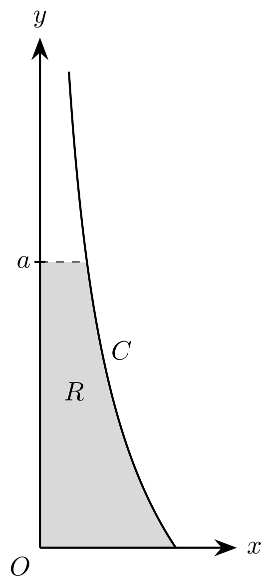
$$\begin{aligned} V &= \pi \left[ \sinh x - x \right]_0^{2\ln 2} \\ &= \pi \left( \sinh(2\ln 2) - 2\ln 2 - (\sinh 0 - 0) \right) \\ &= \pi \left( \sinh(2\ln 2) - 2\ln 2 \right) \end{aligned}$$

From part (a), $\sinh(2\ln 2) = \dfrac{15}{8}$, so

$$V = \pi \left( \frac{15}{8} - 2\ln 2 \right) \text{ cm}^3$$

Therefore, the exact volume of the ornament is

$$\pi \left( \frac{15}{8} - 2\ln 2 \right) \text{ cm}^3$$




**6.** The curve $C$ is given parametrically by

$$x = \frac{9}{t^4}, \qquad y = t^2 - 3, \qquad t \geq \sqrt{3}$$

The shaded region $R$, shown in the diagram, is enclosed by $C$, the $x$-axis, the $y$-axis and the line $y = a$. When $R$ is rotated through $2\pi$ radians about the $y$-axis, the resulting solid has volume

$$\frac{7\pi}{8}$$

Find the exact value of $a$. **[8]**

**Solution**

First write the curve in the form $x$ as a function of $y$.
From

$$y = t^2 - 3$$

we have

$$t^2 = y + 3$$

So

$$x = \frac{9}{t^4} = \frac{9}{(t^2)^2} = \frac{9}{(y+3)^2}$$

Since $t \geq \sqrt{3}$, we have $y \geq 0$, so the region runs from $y = 0$ up to $y = a$.
When the region is rotated about the $y$-axis, the volume is

$$V = \pi \int_0^a x^2 \, \mathrm{d}y$$

Hence

$$\begin{aligned}
V &= \pi \int_0^a \left( \frac{9}{(y+3)^2} \right)^2 \, \mathrm{d}y \\
&= \pi \int_0^a \frac{81}{(y+3)^4} \, \mathrm{d}y \\
&= 81\pi \int_0^a (y+3)^{-4} \, \mathrm{d}y \\
&= 81\pi \left[ \frac{(y+3)^{-3}}{-3} \right]_0^a \\
&= 81\pi \left[ -\frac{1}{3(y+3)^3} \right]_0^a \\
&= 81\pi \left( -\frac{1}{3(a+3)^3} + \frac{1}{3 \cdot 3^3} \right) \\
&= 81\pi \left( -\frac{1}{3(a+3)^3} + \frac{1}{81} \right) \\
&= -\frac{27\pi}{(a+3)^3} + \pi \\
&= \pi \left( 1 - \frac{27}{(a+3)^3} \right)
\end{aligned}$$

We are told that the volume is $\dfrac{7\pi}{8}$, so

$$\begin{aligned}
\pi \left( 1 - \frac{27}{(a+3)^3} \right) &= \frac{7\pi}{8} \\
1 - \frac{27}{(a+3)^3} &= \frac{7}{8} \\
\frac{27}{(a+3)^3} &= \frac{1}{8} \\
(a+3)^3 &= 216
\end{aligned}$$

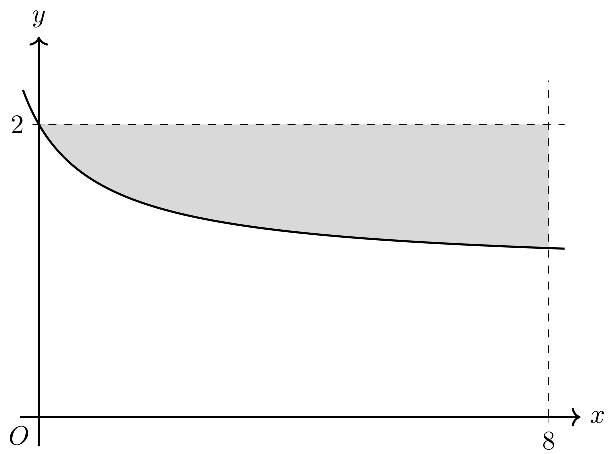
Now $a \geq 0$, so $a + 3 > 0$, and therefore

$$a + 3 = 6$$

Thus

$$a = 3$$

The exact value of $a$ is 3.




**7.** The diagram shows the region bounded by the curve $y = \sqrt{\dfrac{x+4}{x+1}}$, the line $y = 2$ and the line $x = 8$. The curve meets the line $y = 2$ at the point $(0, 2)$.

Find the exact volume of the solid formed when this region is rotated through 360° about the $x$-axis. **[6]**

**Solution**

The region is between the line $y = 2$ and the curve

$$y = \sqrt{\frac{x+4}{x+1}}$$

from $x = 0$ to $x = 8$.
When this region is rotated about the $x$-axis, the volume is

$$V = \pi \int_0^8 \left(2^2 - \left(\sqrt{\frac{x+4}{x+1}}\right)^2\right) \mathrm{d}x$$

So

$$V = \pi \int_0^8 \left(4 - \frac{x+4}{x+1}\right) \mathrm{d}x$$

Simplify the integrand:

$$\begin{aligned} 4 - \frac{x+4}{x+1} &= \frac{4(x+1) - (x+4)}{x+1} \\ &= \frac{4x + 4 - x - 4}{x+1} \\ &= \frac{3x}{x+1} \end{aligned}$$

Hence

$$V = \pi \int_0^8 \frac{3x}{x+1} \mathrm{d}x = 3\pi \int_0^8 \frac{x}{x+1} \mathrm{d}x$$

Now write

$$\frac{x}{x+1} = 1 - \frac{1}{x+1}$$

Therefore

$$V = 3\pi \int_0^8 \left(1 - \frac{1}{x+1}\right) \mathrm{d}x$$

Integrating gives

$$V = 3\pi \left[x - \ln(x+1)\right]_0^8$$

Substitute the limits:

$$\begin{aligned} V &= 3\pi \left((8 - \ln 9) - (0 - \ln 1)\right) \\ &= 3\pi(8 - \ln 9) \end{aligned}$$

Since $\ln 9 = 2\ln 3$,

$$\begin{aligned} V &= 24\pi - 3\pi \ln 9 \\ &= 24\pi - 6\pi \ln 3 \\ &= 6\pi(4 - \ln 3) \end{aligned}$$

So the exact volume is $6\pi(4 - \ln 3)$.

**8.** The region in the first quadrant bounded by the curve $y = \sinh x$, the $y$-axis, and the line $y = 2$ is rotated through $360°$ about the $x$-axis.

Find the exact volume of revolution generated, expressing your answer in a form involving a logarithm. **[7]**

**Solution**

Let the curve $y = \sinh x$ meet the line $y = 2$ at $x = a$.
Then

$$\sinh a = 2$$

To find $a$ in logarithmic form,

$$\begin{aligned} \sinh a = 2 &\implies \frac{e^a - e^{-a}}{2} = 2 \\ &\implies e^a - e^{-a} = 4 \\ &\implies e^{2a} - 1 = 4e^a \end{aligned}$$

Let $u = e^a$. Then $u > 0$ and

$$u^2 - 4u - 1 = 0$$

So

$$u = \frac{4 \pm \sqrt{16+4}}{2} = 2 \pm \sqrt{5}$$

Since $u > 0$, we take

$$u = 2 + \sqrt{5}$$

Hence

$$a = \ln(2 + \sqrt{5})$$

For $0 \le x \le a$, the region lies between $y = \sinh x$ and $y = 2$. When it is rotated about the $x$-axis, the volume is

$$V = \pi \int_0^a \left(2^2 - (\sinh x)^2\right)\, \mathrm{d}x$$

So

$$V = \pi \int_0^a \left(4 - \sinh^2 x\right)\, \mathrm{d}x$$

Use

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

Then

$$\begin{aligned} V &= \pi \int_0^a \left(4 - \frac{\cosh 2x - 1}{2}\right) \mathrm{d}x \\ &= \pi \int_0^a \left(\frac{9}{2} - \frac{1}{2}\cosh 2x\right) \mathrm{d}x \\ &= \pi \left[\frac{9}{2}x - \frac{1}{4}\sinh 2x\right]_0^a \end{aligned}$$

Now $\sinh a = 2$, and since $a > 0$,

$$\cosh a = \sqrt{1 + \sinh^2 a} = \sqrt{1+4} = \sqrt{5}$$

Therefore

$$\sinh 2a = 2 \sinh a \cosh a = 2(2)(\sqrt{5}) = 4\sqrt{5}$$

Substituting into the volume,

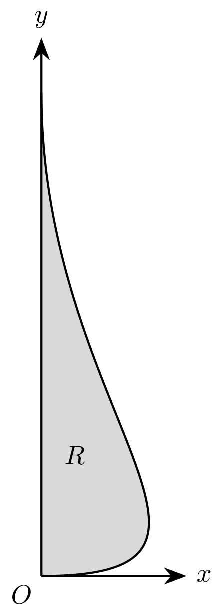
$$\begin{aligned} V &= \pi \left(\frac{9}{2}a - \frac{1}{4}\sinh 2a\right) \\ &= \pi \left(\frac{9}{2}a - \frac{1}{4}(4\sqrt{5})\right) \\ &= \pi \left(\frac{9}{2}a - \sqrt{5}\right) \end{aligned}$$

Finally, using $a = \ln(2+\sqrt{5})$,

$$V = \pi\left(\frac{9}{2}\ln(2+\sqrt{5}) - \sqrt{5}\right)$$

So the exact volume is

$$\pi\left(\frac{9}{2}\ln(2+\sqrt{5}) - \sqrt{5}\right)$$




**9.** Part of the side profile of a solid wooden ornament is shown in the diagram.
The outline is modelled by the curve with parametric equations

$$x = 12t(1-t)^2, \quad y = 8t^2, \qquad 0 \le t \le 1$$

The ornament is formed by rotating the shaded region through $2\pi$ radians about the $y$-axis.

Use the model to find the volume of the ornament. **[7]**

**Solution**

For a solid formed by rotating a region about the $y$-axis, we use

$$V = \pi \int x^2 \, \mathrm{d}y$$

Here

$$x = 12t(1-t)^2, \qquad y = 8t^2, \qquad 0 \le t \le 1$$

Also,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 16t$$

Since $0 \le t \le 1$, we have $\frac{\mathrm{d}y}{\mathrm{d}t} \ge 0$, so $y$ increases through the interval and we can write

$$\begin{aligned} V &= \pi \int_0^1 x^2 \frac{\mathrm{d}y}{\mathrm{d}t} \, \mathrm{d}t \\ &= \pi \int_0^1 \left(12t(1-t)^2\right)^2 (16t) \, \mathrm{d}t \\ &= \pi \int_0^1 144t^2(1-t)^4 \cdot 16t \, \mathrm{d}t \\ &= 2304\pi \int_0^1 t^3(1-t)^4 \, \mathrm{d}t \end{aligned}$$

Now expand

$$(1-t)^4 = 1 - 4t + 6t^2 - 4t^3 + t^4$$

So

$$t^3(1-t)^4 = t^3 - 4t^4 + 6t^5 - 4t^6 + t^7$$

Hence

$$\begin{aligned} V &= 2304\pi \int_0^1 \left(t^3 - 4t^4 + 6t^5 - 4t^6 + t^7\right) \, \mathrm{d}t \\ &= 2304\pi \left[\frac{t^4}{4} - \frac{4t^5}{5} + t^6 - \frac{4t^7}{7} + \frac{t^8}{8}\right]_0^1 \\ &= 2304\pi \left(\frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8}\right) \end{aligned}$$

Using a common denominator of 280,

$$\frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{70 - 224 + 280 - 160 + 35}{280} = \frac{1}{280}$$

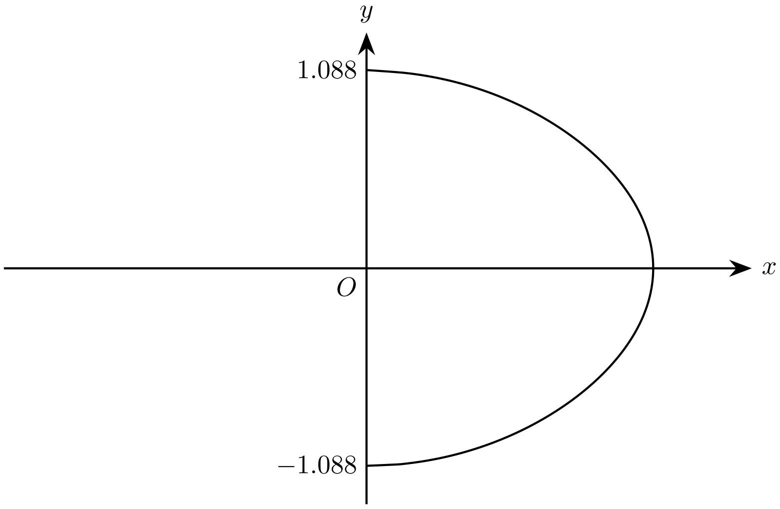
Therefore

$$\begin{aligned} V &= 2304\pi \cdot \frac{1}{280} \\ &= \frac{288\pi}{35} \end{aligned}$$

So the volume of the ornament is

$$\frac{288\pi}{35} \text{ units}^3$$

which is approximately 25.9 units$^3$.




**10.** A chocolatier makes hand-rolled chocolate drops.

The tallest drop in one tray was approximately 2.2 cm high.

The shape of this drop is modelled by rotating the curve with equation

$$20x^2 + 6y^2 + y\sin(2y) = 8, \qquad x \geq 0$$

shown in the diagram above, about the $y$-axis through $2\pi$ radians, where the units are cm.

Given that the $y$-intercepts of the curve are $-1.088$ and $1.088$ to four significant figures,

**(a)** Use algebraic integration to determine, according to the model, the volume of this chocolate drop. **[6]**

The chocolatier melts down 80 chocolate drops from the same tray.

**(b)** Use your answer to part (a) to decide whether, in reality, there is likely to be enough chocolate to fill a mould of volume $140\,\text{cm}^3$, giving a reason. **[2]**

**Solution**

**(a)** At the $y$-intercepts, $x = 0$, so the limits of integration are

$$y = -1.088 \quad \text{and} \quad y = 1.088$$

From

$$20x^2 + 6y^2 + y\sin(2y) = 8$$

we rearrange to get

$$x^2 = \frac{8 - 6y^2 - y\sin(2y)}{20}$$

When this curve is rotated about the $y$-axis, the volume is

$$V = \pi \int_{-1.088}^{1.088} x^2 \,\mathrm{d}y$$

So

$$V = \frac{\pi}{20} \int_{-1.088}^{1.088} \left(8 - 6y^2 - y\sin(2y)\right) \,\mathrm{d}y$$

Using integration by parts for $\int -y\sin(2y)\,\mathrm{d}y$, let

$$u = y, \qquad \frac{\mathrm{d}v}{\mathrm{d}y} = -\sin(2y)$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}y} = 1, \qquad v = \frac{1}{2}\cos(2y)$$

Hence

$$\begin{aligned}\int -y\sin(2y)\,\mathrm{d}y &= uv - \int v\,\frac{\mathrm{d}u}{\mathrm{d}y}\,\mathrm{d}y \\ &= \frac{y}{2}\cos(2y) - \int \frac{1}{2}\cos(2y)\,\mathrm{d}y \\ &= \frac{y}{2}\cos(2y) - \frac{1}{4}\sin(2y)\end{aligned}$$

Therefore

$$\begin{aligned}V &= \frac{\pi}{20}\left[8y - 2y^3 + \frac{y}{2}\cos(2y) - \frac{1}{4}\sin(2y)\right]_{-1.088}^{1.088} \\ &\approx 1.764\ldots\end{aligned}$$

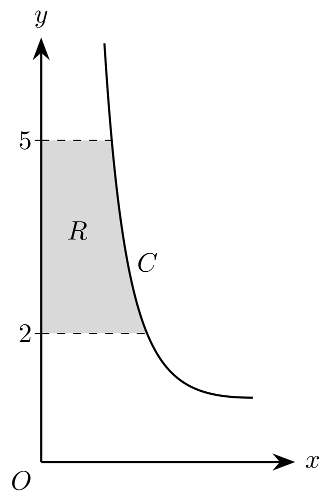
Hence, according to the model, the volume of the chocolate drop is $1.76\,\mathrm{cm}^3$.

**(b)** Using the answer from part (a), the volume of 80 such drops would be estimated as

$$80 \times 1.76 = 140.8\,\mathrm{cm}^3$$

This is only just above $140\,\mathrm{cm}^3$. Also, the model in part (a) is for the tallest drop in the tray, so using $1.76\,\mathrm{cm}^3$ for all 80 drops will overestimate the total amount of chocolate.

Therefore, in reality, there is unlikely to be enough chocolate to fill the $140\,\mathrm{cm}^3$ mould.




**11.** The diagram shows part of the curve $C$ with parametric equations

$$x = \frac{2}{1+t}, \quad y = t^2 + 1, \qquad t \geq 0$$

The region $R$ is bounded by the curve, the $y$-axis and the lines $y = 2$ and $y = 5$. Region $R$ is rotated through $2\pi$ radians about the $y$-axis.

Use parametric integration to find the volume of the resulting solid of revolution. **[6]**

**Solution**

For the given curve,

$$x = \frac{2}{1+t}, \qquad y = t^2 + 1, \qquad t \geq 0$$

First find the values of $t$ corresponding to the horizontal boundaries.
When $y = 2$,

$$t^2 + 1 = 2 \implies t^2 = 1 \implies t = 1$$

since $t \geq 0$.
When $y = 5$,

$$t^2 + 1 = 5 \implies t^2 = 4 \implies t = 2$$

So the required part of the curve corresponds to $1 \leq t \leq 2$.
When the region is rotated about the $y$-axis, the volume is

$$V = \pi \int x^2 \, \mathrm{d}y$$

Using the parameter $t$,

$$V = \pi \int_1^2 x^2 \frac{\mathrm{d}y}{\mathrm{d}t} \, \mathrm{d}t$$

Now

$$x^2 = \left( \frac{2}{1+t} \right)^2 = \frac{4}{(1+t)^2}$$

and

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2t$$

Hence

$$V = \pi \int_1^2 \frac{4}{(1+t)^2} \cdot 2t \, \mathrm{d}t = 8\pi \int_1^2 \frac{t}{(1+t)^2} \, \mathrm{d}t$$

Now simplify the integrand:

$$\frac{t}{(1+t)^2} = \frac{(1+t) - 1}{(1+t)^2} = \frac{1}{1+t} - \frac{1}{(1+t)^2}$$

So

$$V = 8\pi \int_1^2 \left( \frac{1}{1+t} - \frac{1}{(1+t)^2} \right) \mathrm{d}t$$

Integrating,

$$\int \frac{1}{1+t} \, \mathrm{d}t = \ln(1+t)$$

and

$$\int -\frac{1}{(1+t)^2} \, \mathrm{d}t = \frac{1}{1+t}$$

Therefore

$$V = 8\pi \left[ \ln(1+t) + \frac{1}{1+t} \right]_1^2$$

Substituting the limits,

$$V = 8\pi \left( \ln 3 + \frac{1}{3} - \ln 2 - \frac{1}{2} \right)$$

So

$$V = 8\pi \left( \ln \frac{3}{2} - \frac{1}{6} \right)$$

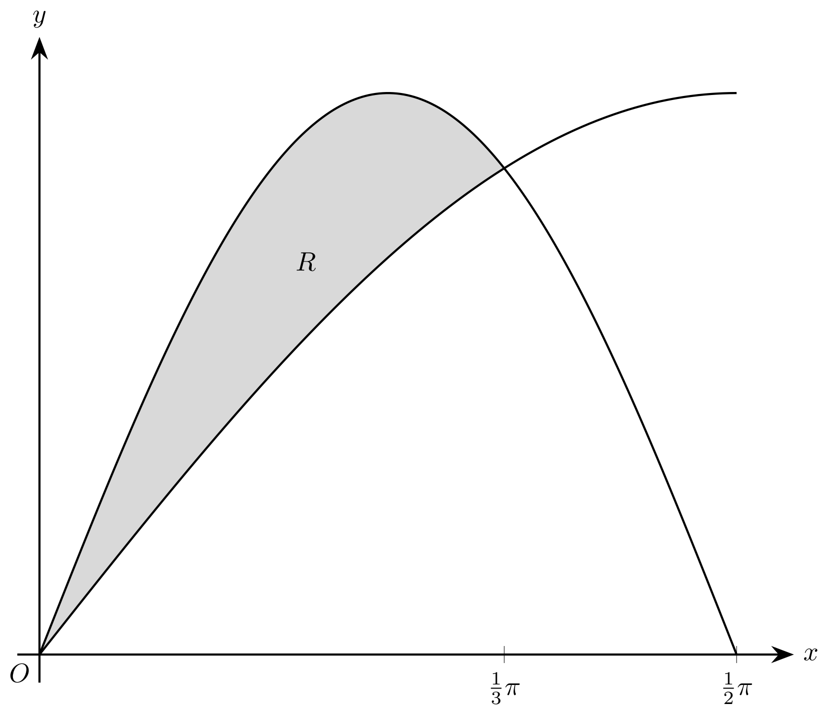
$$V = 8\pi \ln \frac{3}{2} - \frac{4\pi}{3}$$

Hence the volume of the solid is

$$\boxed{8\pi \ln \frac{3}{2} - \frac{4\pi}{3}}$$

In decimal form,

$$V \approx 6.00$$




**12.** The diagram shows the curves $y = \sin x$ and $y = \sin 2x$, for $0 \leq x \leq \frac{1}{2}\pi$. The shaded region $R$ is the finite region enclosed by the curves.

Find the volume of the solid of revolution formed when $R$ is rotated completely about the $x$-axis, giving your answer in terms of $\pi$. **[7]**

**Solution**

First find where the curves meet.

$$\sin x = \sin 2x$$

Using $\sin 2x = 2 \sin x \cos x$,

$$\sin x = 2 \sin x \cos x$$

$$\sin x(2 \cos x - 1) = 0$$

So, for $0 \leq x \leq \frac{\pi}{2}$,

$$\sin x = 0 \Rightarrow x = 0 \qquad \text{or} \qquad 2 \cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$$

Hence the shaded region is between $x = 0$ and $x = \frac{\pi}{3}$.
For $0 < x < \frac{\pi}{3}$, we have $\sin x > 0$ and $\cos x > \frac{1}{2}$, so $2 \cos x - 1 > 0$. Therefore

$$\sin x(2 \cos x - 1) > 0$$

which means

$$\sin 2x > \sin x$$

So when the region is rotated about the $x$-axis, the volume is

$$V = \pi \int_0^{\pi/3} \left( (\sin 2x)^2 - (\sin x)^2 \right) \, \mathrm{d}x$$

Now use

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

Then

$$\begin{aligned}\sin^2 2x - \sin^2 x &= \frac{1-\cos 4x}{2} - \frac{1-\cos 2x}{2}\\ &= \frac{\cos 2x - \cos 4x}{2}\end{aligned}$$

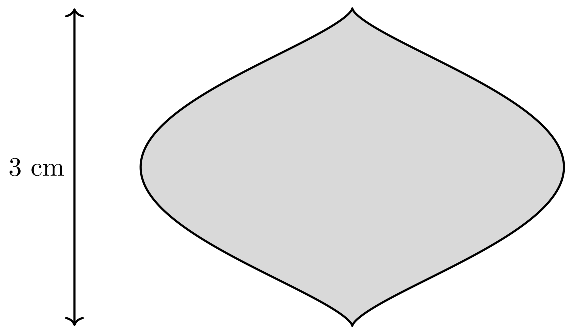
So

$$\begin{aligned}V &= \pi\int_0^{\pi/3} \frac{\cos 2x - \cos 4x}{2}\,\mathrm{d}x\\ &= \pi\left[\frac{\sin 2x}{4} - \frac{\sin 4x}{8}\right]_0^{\pi/3}\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}V &= \pi\left(\frac{\sin(2\pi/3)}{4} - \frac{\sin(4\pi/3)}{8}\right)\\ &= \pi\left(\frac{\sqrt{3}/2}{4} - \frac{-\sqrt{3}/2}{8}\right)\\ &= \pi\left(\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{16}\right)\\ &= \frac{3\sqrt{3}\pi}{16}\end{aligned}$$

The volume of the solid of revolution is $\dfrac{3\sqrt{3}\pi}{16}$.




**13.** The diagram shows the outline of a glass bead. The bead is modelled by the solid obtained when the region enclosed by a curve $C$ is rotated about the $y$-axis. The actual bead has height 3 cm. The curve $C$ has parametric equations

$$x = \sin\theta - \frac{1}{3}\sin 3\theta, \qquad y = 1 - \cos\theta, \qquad 0 \leq \theta \leq 2\pi$$

**(a)** Show that a Cartesian equation of the curve $C$ is

$$x^2 = \frac{16}{9}y^3(2-y)^3$$

**[4]**

**(b)** Hence, using the model, find, in $\text{cm}^3$, the volume of the bead. **[5]**

**Solution**

**(a)** Using the identity

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

we have

$$\begin{aligned} x &= \sin\theta - \frac{1}{3}\sin 3\theta \\ &= \sin\theta - \frac{1}{3}\left(3\sin\theta - 4\sin^3\theta\right) \\ &= \sin\theta - \sin\theta + \frac{4}{3}\sin^3\theta \\ &= \frac{4}{3}\sin^3\theta \end{aligned}$$

So

$$x^2 = \frac{16}{9}\sin^6\theta$$

Also

$$y = 1 - \cos\theta$$

so

$$\cos\theta = 1 - y$$

Hence

$$\begin{aligned} \sin^2\theta &= 1 - \cos^2\theta \\ &= 1 - (1-y)^2 \\ &= 1 - (1 - 2y + y^2) \\ &= 2y - y^2 \\ &= y(2-y) \end{aligned}$$

Therefore

$$\sin^6\theta = \left(\sin^2\theta\right)^3 = (y(2-y))^3$$

and so

$$\begin{aligned} x^2 &= \frac{16}{9}\sin^6\theta \\ &= \frac{16}{9}\left(y(2-y)\right)^3 \\ &= \frac{16}{9}y^3(2-y)^3 \end{aligned}$$

Thus a Cartesian equation of $C$ is

$$x^2 = \frac{16}{9}y^3(2-y)^3$$

**(b)** The model has height

$$2 - 0 = 2$$

whereas the actual bead has height 3 cm, so the linear scale factor is

$$k = \frac{3}{2}$$

First find the volume of the model.
When the region is rotated about the $y$-axis,

$$V_{\text{model}} = \pi\int_0^2 x^2\,\mathrm{d}y$$

Using the result from part (a),

$$\begin{aligned} V_{\text{model}} &= \pi\int_0^2 \frac{16}{9}y^3(2-y)^3\,\mathrm{d}y \\ &= \frac{16\pi}{9}\int_0^2 y^3(2-y)^3\,\mathrm{d}y \end{aligned}$$

Now expand:

$$\begin{aligned} (2-y)^3 &= 8 - 12y + 6y^2 - y^3 \\ y^3(2-y)^3 &= 8y^3 - 12y^4 + 6y^5 - y^6 \end{aligned}$$

So

$$\begin{aligned} V_{\text{model}} &= \frac{16\pi}{9}\int_0^2 \left(8y^3 - 12y^4 + 6y^5 - y^6\right)\,\mathrm{d}y \\ &= \frac{16\pi}{9}\left[2y^4 - \frac{12}{5}y^5 + y^6 - \frac{1}{7}y^7\right]_0^2 \end{aligned}$$

Substituting the limits,

$$\begin{aligned} V_{\text{model}} &= \frac{16\pi}{9}\left(2(2^4) - \frac{12}{5}(2^5) + 2^6 - \frac{1}{7}(2^7)\right) \\ &= \frac{16\pi}{9}\left(32 - \frac{384}{5} + 64 - \frac{128}{7}\right) \\ &= \frac{16\pi}{9}\cdot\frac{32}{35} \\ &= \frac{512\pi}{315} \end{aligned}$$

Volume scales as the cube of the linear scale factor, so

$$\begin{aligned} V_{\text{actual}} &= \left(\frac{3}{2}\right)^3 V_{\text{model}} \\ &= \frac{27}{8}\times\frac{512\pi}{315} \\ &= \frac{192\pi}{35} \end{aligned}$$

Therefore the volume of the bead is

$$\frac{192\pi}{35}\ \text{cm}^3 \approx 17.2\ \text{cm}^3$$

**14.** The equation of a curve is

$$y = \frac{x}{\sqrt{k^2 + x^2}}$$

where $k$ is a positive constant.

The region in the first quadrant bounded by the curve, the $x$-axis, the $y$-axis and the line $x = k$ is rotated through $2\pi$ radians about the $x$-axis
Given that the volume of revolution formed is 1 unit$^3$, find the exact value of $k$ **[4]**

**Solution**

When the region is rotated about the $x$-axis, the volume is

$$V = \pi \int_0^k y^2 \, \mathrm{d}x$$

Here

$$y = \frac{x}{\sqrt{k^2 + x^2}} \qquad \Rightarrow \qquad y^2 = \frac{x^2}{k^2 + x^2}$$

So

$$V = \pi \int_0^k \frac{x^2}{k^2 + x^2} \, \mathrm{d}x$$

Rewrite the integrand:

$$\frac{x^2}{k^2 + x^2} = \frac{k^2 + x^2 - k^2}{k^2 + x^2} = 1 - \frac{k^2}{k^2 + x^2}$$

Hence

$$V = \pi \int_0^k \left(1 - \frac{k^2}{k^2 + x^2}\right) \mathrm{d}x$$

Now integrate:

$$\begin{aligned} \int \left(1 - \frac{k^2}{k^2 + x^2}\right) \mathrm{d}x &= x - k^2 \int \frac{1}{k^2 + x^2} \, \mathrm{d}x \\ &= x - k^2 \left(\frac{1}{k} \arctan \frac{x}{k}\right) \\ &= x - k \arctan \frac{x}{k} \end{aligned}$$

Therefore

$$\begin{aligned} V &= \pi \left[x - k \arctan \frac{x}{k}\right]_0^k \\ &= \pi \left(k - k \arctan 1\right) \\ &= \pi \left(k - k \cdot \frac{\pi}{4}\right) \\ &= \pi k \left(1 - \frac{\pi}{4}\right) \end{aligned}$$

We are given that the volume is 1, so

$$\pi k \left(1 - \frac{\pi}{4}\right) = 1$$

Solving for $k$:

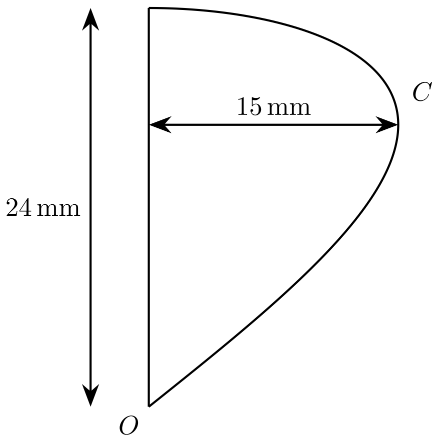
$$\begin{aligned} k &= \frac{1}{\pi \left(1 - \frac{\pi}{4}\right)} \\ &= \frac{1}{\pi \left(\frac{4 - \pi}{4}\right)} \\ &= \frac{4}{\pi(4 - \pi)} \end{aligned}$$

Since $k$ is positive, this is valid.
The exact value of $k$ is

$$k = \frac{4}{\pi(4 - \pi)}$$

**15. (a)** Prove that

$$\cos 4\theta \cos\theta \equiv \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$ **[3]**




The diagram shows the cross-section of a decorative ornament.

The ornament can be modelled by rotating the region enclosed by the curve $C$ and the $y$-axis about the $y$-axis. Curve $C$ has parametric equations

$$x = 15\sin 2\theta, \quad y = 24\sin\theta, \quad 0 \le \theta \le \frac{\pi}{2}$$

where $x$ and $y$ are measured in millimetres.

**(b)** Find the volume of the ornament. **[5]**

**Solution**

**(a)** Using the cosine addition formula,

$$\cos 5\theta = \cos(4\theta + \theta) = \cos 4\theta \cos\theta - \sin 4\theta \sin\theta$$

and

$$\cos 3\theta = \cos(4\theta - \theta) = \cos 4\theta \cos\theta + \sin 4\theta \sin\theta$$

Adding these gives

$$\begin{aligned}\cos 5\theta + \cos 3\theta &= (\cos 4\theta \cos\theta - \sin 4\theta \sin\theta) + (\cos 4\theta \cos\theta + \sin 4\theta \sin\theta)\\ &= 2\cos 4\theta \cos\theta\end{aligned}$$

Therefore

$$\cos 4\theta \cos\theta \equiv \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$

So the required identity is proved.

**(b)** When the region is rotated about the $y$-axis, the volume is

$$V = \pi \int x^2 \,\mathrm{d}y$$

Using the parameter $\theta$,

$$V = \pi \int_0^{\pi/2} x^2 \frac{\mathrm{d}y}{\mathrm{d}\theta}\,\mathrm{d}\theta$$

Now

$$x = 15\sin 2\theta, \qquad y = 24\sin\theta$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 24\cos\theta$$

Substituting,

$$\begin{aligned} V &= \pi \int_0^{\pi/2} (15\sin 2\theta)^2 (24\cos\theta)\,\mathrm{d}\theta \\ &= 5400\pi \int_0^{\pi/2} \sin^2 2\theta \cos\theta\,\mathrm{d}\theta \end{aligned}$$

Use

$$\sin^2 2\theta = \frac{1-\cos 4\theta}{2}$$

Then

$$\begin{aligned} V &= 5400\pi \int_0^{\pi/2} \frac{1-\cos 4\theta}{2} \cos\theta\,\mathrm{d}\theta \\ &= 2700\pi \int_0^{\pi/2} (1-\cos 4\theta)\cos\theta\,\mathrm{d}\theta \\ &= 2700\pi \left( \int_0^{\pi/2} \cos\theta\,\mathrm{d}\theta - \int_0^{\pi/2} \cos 4\theta \cos\theta\,\mathrm{d}\theta \right) \end{aligned}$$

From part (a),

$$\cos 4\theta \cos\theta = \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$

So

$$\begin{aligned} V &= 2700\pi \int_0^{\pi/2} \left( \cos\theta - \frac{1}{2}(\cos 5\theta + \cos 3\theta) \right)\,\mathrm{d}\theta \\ &= 2700\pi \left[ \sin\theta - \frac{1}{10}\sin 5\theta - \frac{1}{6}\sin 3\theta \right]_0^{\pi/2} \end{aligned}$$

Now evaluate the bracket:

$$\sin\frac{\pi}{2} = 1, \qquad \sin\frac{5\pi}{2} = 1, \qquad \sin\frac{3\pi}{2} = -1$$

Hence

$$\begin{aligned} V &= 2700\pi \left( 1 - \frac{1}{10} - \frac{1}{6}(-1) \right) \\ &= 2700\pi \left( 1 - \frac{1}{10} + \frac{1}{6} \right) \\ &= 2700\pi \left( \frac{16}{15} \right) \\ &= 2880\pi \end{aligned}$$

Therefore the volume of the ornament is

$$2880\pi \text{ mm}^3$$

which is approximately

$$9.05 \times 10^3 \text{ mm}^3$$